PLANET · 03 ENERGY EFFICIENCY

Apex PdM

Hear it before it breaks.

A ~RM 170 sensor pod that feels, hears and meters township machines, so wear and wasted power are caught before a breakdown.

TEAM APEX PDM

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LIVE DASHBOARD

washer-pdm.pages.dev

Code: github.com/adamdanishj30/washer-pdm



Our rig: the pod on top of a Samsung front-load washer

01 · PROBLEM

Township machines are fixed only after they break

>55%

of an average facility's maintenance is still reactive: **run it till it breaks.**

US DOE FEMP, O&M Best Practices Guide, Release 2.0 (2004), §5.2. US figures, not Malaysian or township data.

1. **A small part wears**
   A bearing, damper or seal worth tens of ringgit
2. **The machine works harder**
   It shakes more, sounds different and draws more power
3. **It breaks down**
   No cooling for days; an emergency call-out
4. **It gets replaced**
   A new motor or compressor costs thousands of ringgit

When an AC unit fails, the first to find out are the people sitting under it, often at Friday prayers, Ramadan nights or weddings.

Spotting wear early needs industrial vibration sensors that cost thousands of ringgit each, so only big factories use them.

02 · TARGET USER

Built for the teams that keep shared spaces running

PRIMARY USER

Facility managers

Mosques, suraus, kuils, malls, clubhouses

NEEDS

To know which AC unit, pump or fan needs a look before it fails

GETS

One dashboard for every machine; repairs planned outside busy times

PARTNER

Maintenance contractors

Local firms that service township sites

NEEDS

Fewer emergency call-outs and clearer job lists

GETS

Which machine to check, with the data behind the call

WHO BENEFITS

Residents and visitors

People using prayer halls, malls, clubhouses

NEEDS

Shared spaces that stay cool and working when they are busiest

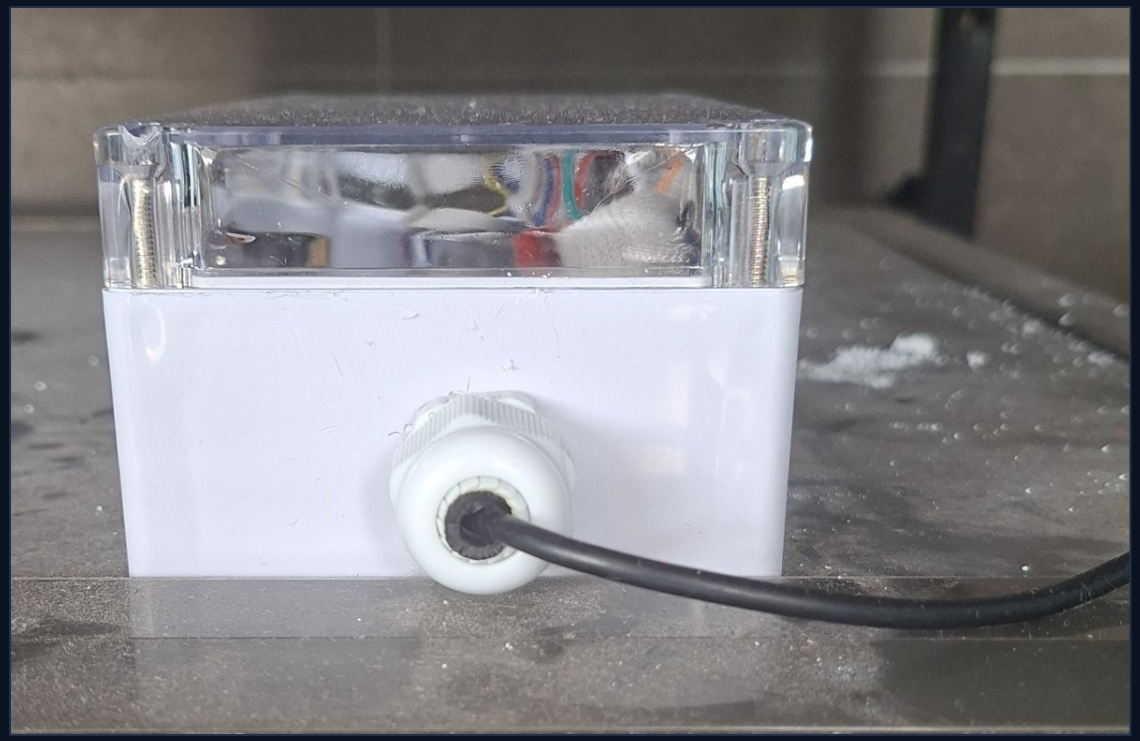
GETS

Fewer breakdowns, less wasted power and less scrap

Why them: they look after many spinning machines (AC compressors, water pumps, fans), but still check them one by one, by hand, and cannot see wear happening inside.

03 · SOLUTION

A pod that feels, hears and meters each machine



The pod sits outside the machine on 3M VHB tape.
Nothing inside is opened; the smart plug needs no rewiring.

FEEL — **Vibration**
MPU-6050 accelerometer, 200 Hz: shaking and drum speed

HEAR — **Sound**
INMP441 microphone, 16 kHz: grinding and rubbing, turned into numbers on the chip, never recorded

METER — **Power**
Tapo P110 smart plug: how hard the motor is working

LEARN — **Each machine's own normal**
Trained on healthy runs only; no broken-machine examples

SHOW — **Live dashboard**
washer-pdm.pages.dev, open in any browser

> Proven first on a front-load washer: a hard test that shakes, sloshes and keeps changing speed.

Three senses, one cheap pod, no fault data needed

	APEX PDM	SINGLE VIBRATION SENSOR	MANUAL INSPECTION
What it senses	Vibration, sound and power, fused	Vibration only	Eyes and ears, on visit days
Cost	About RM 170 per pilot kit	Thousands of ringgit each	Technician time per visit
Fault data needed	None: learns each machine's normal	Generic alarm limits	Relies on experience
Sees wasted power	Yes: power metered directly	No	No
How often it checks	Every 5 seconds, all day	Continuous	When someone visits

Private: no recordings leave the pod; sound becomes numbers on chip

One-class AI: trained only on healthy runs, starts on day one

Non-invasive: sticks on the outside and plugs in, no rewiring

05 · HOW IT WORKS

From raw signals to an alert, every 5 seconds

1 · SENSE

Sensor pod

MPU-6050 vibration, 200 Hz
INMP441 sound, 16 kHz
Tapo P110 power, every 1 s

2 · EDGE

ESP32, two cores

Core 1: sensors + maths
Core 0: Wi-Fi + upload
~4 min buffer if Wi-Fi drops

3 · CLOUD

Firebase database

1 record every 5.024 s
32 signal features
Numbers only, no audio

4 · SEE

Live dashboard

washer-pdm.pages.dev
Vibration, sound, power
Polls every 30 s or more

WHY ON THE CHIP?

Less data to send, lower cloud cost, and no sound recordings ever leave the building.

5 · LEARN

One-class anomaly model (not generative AI)

Fuses vibration (13), sound (15) and power (4) features
Compared Mahalanobis, Isolation Forest and One-Class SVM
Tested leave-one-wash-out: never scored on its own training data

6 · ACT

Flag the machine

The technician is told which unit needs a look, before it breaks

Firmware 07 · FreeRTOS dual-core · I2C + I2S DMA · Firebase Realtime Database · static web dashboard · Python ML pipeline

Catch machines that waste power before they fail

OUR ESTIMATE · TO BE CONFIRMED IN THE PILOT

≈ RM 900 a year

≈ 1,825 kWh a year from one machine with a 5 kW power draw using 10% more because of a worn part.

Assumes 10 h a day and about RM 0.50/kWh. The 10% is an assumption, inside the US DOE guide's 5–20% energy savings from O&M (§2.4).

Payback: a ~RM 170 pod pays for itself within months (estimate).

57% of energy in Malaysian office buildings goes to air-conditioning

Saidur (2009), Energy Policy 37(10), 4104–4113.

70–75% of breakdowns eliminated by predictive maintenance

Surveys cited by the US DOE FEMP O&M Guide R2.0, §5.4 (2004). US figures.

SECOND BENEFIT

Repair, not replace

Fix a small part early: machines last longer, less scrap.

Our smart plug meters each machine, so a unit that starts drawing more power shows up before it breaks.

Every ringgit and kWh figure here is an estimate. No savings have been measured yet; the pilot will measure them.

It already works: a live rig on a real machine



Live dashboard, 21 Oct wash: vibration and sound across one 32-minute run (spin highlighted).

PILOT KIT, NET PAID (RECEIPTS)

RM 169.68

Sensor node RM 131.07 + Tapo P110 plug
Sticks on with VHB tape and plugs in:
no rewiring, nothing inside opened

11 washes

8,379 rows of data recorded
(as of 20 Oct)

>20×

vibration jump from idle
to spin, clearly measured

19.9 Hz

≈ 1,200 rpm spin, seen in
every firmware-07 wash

0 of 8

healthy washes falsely flagged
in a leave-one-wash-out test;
simulated sound fault caught
in 7 of 8. Small sample.

TRL 4–5. Honest limits: one test machine so far, and added vibration was not yet detected in the first test.

From one washer to a township fleet

DONE

Built

- Working pod on a washer
- Live dashboard online
- 11 washes recorded
- First leave-one-wash-out test run

NEXT

Prove the model

- 20+ more washes on firmware 07
- Train and test the new model
- Noise baselines
- Try an air compressor and a water pump

THEN

Township pilot

- About ten AC units at one township site
- Measure real kWh and false-alarm rates
- Check the payback estimate

LATER

Scale up

- Custom board and an industrial accelerometer
- One fleet dashboard for the whole township
- Clubhouses, laundry hubs, booster pumps

At fleet scale: identical units are compared with their neighbours, so the one that starts drawing more power or shaking more stands out (proposed).

TEAM AND ASK

Our ask: one township site for a pilot

WHAT WE NEED

One site to pilot on, e.g. City of Elmina

About ten AC units at one township site. Each pod sticks on outside the unit, with no rewiring.

WHAT THE SITE GETS

Early warning on worn machines, a live dashboard, and measured kWh to check our RM 900 estimate.

A ~RM 170 pod per unit; payback within months (estimate).

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SEE IT LIVE
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Firmware 07 and the first model test

ON THE ESP32 (FIRMWARE 07)

Vibration: MPU-6050 at 200 Hz via FIFO, 44 Hz anti-alias filter (drum 1x/2x orders, rocking)

Sound: INMP441 at 16 kHz over I2S DMA; 2 kHz high-pass RMS and kurtosis computed on chip

Power: Tapo P110 read locally (KLAP), every 1 s

Window: 5.024 s → one 45-column row with 32 signal features (audio 15, vibration 13, power 4)

Cores: Core 1 sensing + DSP; Core 0 Tapo, session gating, batched upload, web server, OTA

Resilience: ~4 min buffer during Wi-Fi drops

Cloud budget: dashboard polls every 30 s or slower; only features are uploaded

Wash programmes differ (quick, long cold, long hot), inferred from power.

NESTED LEAVE-ONE-WASH-OUT TEST (8 EARLIER HEALTHY WASHES)

5 s scores → out-of-fold percentiles → slow (~10 min) average → wash flag

Healthy washes falsely flagged Mahalanobis and Isolation Forest; 95% upper bound 31%	**0 of 8**
Simulated sound fault (2× loudness) Smaller sound changes were mostly missed	**7 of 8**
Simulated added vibration Up to 0.2 m/s² added RMS	**Not detected**

Why the limit: these 8 washes use the older format (one 5 s snapshot, no phase labels), so impulsive and spectral features could not be computed. Firmware 07 fixes this; its model needs at least 3 washes per type.

Source: ml/models/loco_report.md (simulated faults, not real ones).

How we got about RM 900 a year (our estimate)

One machine with a 5 kW power draw × 10 h a day × 365 days	**≈ 18,250 kWh a year**
A worn part makes it draw 10% more (assumption)	**≈ 1,825 kWh a year**
× about RM 0.50 per kWh	**≈ RM 900 a year**
A ~RM 170 pod that catches one such problem	**pays for itself within months**

All four lines are estimates to be measured in the pilot. The 10% sits inside the 5–20% O&M energy savings in the US DOE guide (§2.4).

REFERENCES

[1] US Department of Energy, Federal Energy Management Program. Operations & Maintenance Best Practices: A Guide to Achieving Operational Efficiency, Release 2.0 (PNNL-14788, July 2004), §2.4, §5.2, §5.4. US figures; the reactive share comes from a 2000 study.

[2] Saidur, R. (2009). Energy consumption, energy savings, and emission analysis in Malaysian office buildings. Energy Policy, 37(10), 4104–4113. doi:10.1016/j.enpol.2009.04.052

[3] Melgaard, S. P., et al. (2022). Fault Detection and Diagnosis Encyclopedia for Building Systems: A Systematic Review. Energies, 15(12), 4366. doi:10.3390/en15124366

Our own evaluation: ml/models/loco_report.md at github.com/adamdanishj30/washer-pdm